Bryan Wei

LinkedIn GitHub Portfolio l76wei@uwaterloo.ca

Skills

Languages: Python, TypeScript, JavaScript, Java, C++, C, SQL, Bash, PowerShell

Frameworks: React, Next.js, React Native, three.js, Node.js, Express, FastAPI, Spring Boot, PyTorch, Axios

Tools & Databases: Git, Linux, AWS, Azure, Unity, Bun, PostgreSQL, MySQL, SQLite, MongoDB, Convex, Tinybird

Education

University of Waterloo, *Honours Computer Science Co-op* *Sep 2024 - Present*

Experience

Full Stack Developer *Sep 2026 - Present*
Workplace Safety and Insurance Board - Innovation Lab *Waterloo, ON*

- Engineered a **test data generation engine** that writes customizable data into **SQL** files or **Microsoft Fabric** tables, cutting generation time for **1 million rows** from **20s to 7s** with **multithreading**

Software Developer *Sep 2026 - Present*
Waterloo Reality Labs *Waterloo, ON*

- Engineered a **gesture intensity scoring module** that rates how much effort goes into any hand gesture on a universal 0–1 scale, achieving ≥**90% ranking accuracy** across effort levels by combining joint speed, snap, and move time
- Developed a **few-shot Siamese network** that recognizes similar hand gestures based on 1–3 example clips with up to **98% accuracy**, using **PyTorch triplet loss** and exporting to **ONNX** for **Unity**

Software Developer *May 2026 - Aug 2026*
Traces *Toronto, ON*

- Contributed to Traces, a developer platform used by **100+ organizations** for capturing, sharing, and continuing coding sessions across different coding agents and teams
- Built and maintained **data ingestion adapters** for **14+ coding agents**, including local session discovery, content parsing, and **schema normalization** across **JSONL** files, **SQLite** databases, and cloud-backed session stores
- Built a **subagent discovery pipeline** that infers subagent relationships for **8+ agents** with **97% accuracy**, preserving lineage across local indexing, API surfaces, remote sync, and **Convex** storage
- Developed a **token usage tracking** feature for **8+ agents**, mapping agent-specific per-API-call and session-level usage into shared **TypeScript** schemas, with CLI sync and persistence in **Convex** and **Tinybird**, achieving **99% accuracy**
- Designed a provider-neutral **cloud agent session capture engine** that works across polling, webhook, and streaming interfaces, with **concurrent job processing** and **idempotent, retry-safe persistence**

Projects

StickDrift *Oct 2026*

- Developed a **movement disorder monitoring platform** for seniors using **React**, **TypeScript**, **three.js**, **FastAPI**, and **PostgreSQL**, where controller games capture joystick and accelerometer data to detect early signs of movement disorders with **machine learning**
- Trained a **PyTorch TCN** to identify hand-movement patterns associated with tremor, stroke, and Parkinson's, integrating pattern scores into a caregiver dashboard

Contour *Sep 2026*

- Built a **project visualization platform** using **React**, **TypeScript**, **Node.js**, **Express**, and **PostgreSQL**, automatically transforming GitHub activity and coding-agent sessions into an interactive project map
- Engineered **GitHub webhook** ingestion for pull requests, reviews, and CI runs, plus a local collector that scans **Codex**, **Claude Code**, and **Cursor** sessions and uses **SQLite** checkpoints to **incrementally sync** new activity
- Designed an **asynchronous event processing pipeline** using **pg-boss** and **PostgreSQL**, integrating the **OpenAI API** to organize development activity into the live map